VERAPAMIL

Never with IV β-blocker

Constipation common

CHEMICAL NAME (RS)-2-(3,4-dimethoxyphenyl)-5-{[2-(3,4-dimethoxyphenyl)ethyl](methyl)amino}-2-isopropylpentanenitrile (hydrochloride)

CLASS Non-dihydropyridine calcium-channel blocker (phenylalkylamine); class IV antiarrhythmic

TRADE NAMES Isoptin · Calan · Securon · Univer

THERAPEUTIC USES / INDICATIONS

- Supraventricular tachycardia (SVT) termination (IV) and prevention.
- AF/flutter rate control.
- Angina; hypertension.
- Cluster headache prophylaxis (high dose).

ADMINISTRATION

- **PO** IR or MR (swallow whole).
- **IV:** 5–10 mg over at least 2 min (3 min in elderly) with ECG monitoring.
- Have atropine, calcium and pacing available.

DOSAGES

- **SVT (IV):** 5–10 mg over 2 min; repeat 5 mg after 5–10 min if needed.
- **Angina:** 80–120 mg three times daily.
- **Hypertension:** MR 120–480 mg daily.
- **Cluster headache prevention (off-label):** specialist titration with ECG monitoring.

PHARMACOKINETICS

- **Onset:** PO 1–2 h; IV 1–5 min. **Duration:** 6–8 h (IR).
- Extensive first-pass metabolism (CYP3A4); inhibits CYP3A4 and P-gp.
- Excreted in urine. **t½:** 3–7 h.

PHARMACODYNAMICS

- Blocks **L-type calcium channels** → slows AV node conduction and SA node rate; strong negative inotrope.
- Vasodilates coronary and peripheral arteries.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Constipation.
- Bradycardia, dizziness, headache.
- Ankle oedema, flushing.

ADVERSE EFFECTS (SERIOUS)

- AV block, asystole (especially with β-blockers).
- Hypotension.
- Heart failure.
- VF in WPW with AF or in VT misdiagnosed as SVT.

CONTRAINDICATIONS & CAUTIONS

- IV verapamil with IV β-blocker (asystole).
- 2nd/3rd-degree AV block, sick sinus syndrome, severe bradycardia.
- HFrEF, cardiogenic shock, hypotension.
- AF/flutter with WPW; broad-complex tachycardia of unknown origin.

KEY INTERACTIONS

- **β-blockers:** severe bradycardia, AV block.
- **Digoxin:** ↑ digoxin levels (reduce dose).
- **Simvastatin, colchicine, ciclosporin, dabigatran:** ↑ levels.
- **Grapefruit:** ↑ verapamil.

NURSING MANAGEMENT

- Confirm narrow-complex SVT before IV use; continuous ECG.
- Check HR and BP before oral doses.
- Monitor bowel habit; give fibre/laxative.
- Watch for oedema, breathlessness.
- Check digoxin levels if co-prescribed.

PATIENT EDUCATION

- Increase fibre and fluids for constipation.
- Report slow pulse, fainting, breathlessness.
- Avoid grapefruit juice.
- Swallow MR tablets whole.

KEY POINTS

- Class IV antiarrhythmic — slows AV node.
- Never with IV β-blocker; avoid in WPW-AF.
- SVT: 5–10 mg IV over 2 min.
- Constipation; raises digoxin.

SOURCES CHECKED **[322]** Verapamil 40mg/5ml Oral Solution (UK SmPC) **[323]** Verapamil 2.5 mg/ml Solution for injection (UK SmPC) **[324]** Verapamil Hydrochloride, oral (US label)

Full references p. 432–457